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PTx_protocols

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PTX_DL_SAG_MP2RAGE_EDGE_CS_DL_0.63MM_Q9WS105
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PTX_SAG_DIR_FLAIR_SPC_Q9WS104-0.78mm

PTX_SAG_T2prep_FLAIR_SPC_Q9WS104-0.78mm

PTX_SAG_T2_SPC_0.6_Q9WS104

PTX DL SAG MP2RAGE EDGE CS DL

0.63MM_Q9WS105

COR	T2	TSE	HIPPO	DR	+PSS
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\\USER\Renzo\PTx_protocols\copied\PTX_DL_SAG_MP2RAGE_EDGE_CS_DL_0.63MM_Q9WS105

TA: 6:51 min Coil Selection: Auto Voxel Size: 0.6×0.6×0.6 mm³ Acc:: 7.3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	On
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	1st Segment
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A12.2 F22.8 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	256
Phase Oversampling	0 %
Slice Oversampling	12.5 %
FOV Read	240 mm
FOV Phase	93.8 %
Slice Thickness	0.63 mm
TR	6000.0 ms
TE	3.02 ms
Averages	1
Concatenations	1
AutoAlign	Head > Brain

Contrast - Common

TR	6000.0 ms
TE	3.02 ms
Magn. Preparation	Non-sel. IR
TI 1	875 ms
TI 2	2700 ms
Flip Angle 1	4 deg
Flip Angle 2	5 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Contrast - Dynamic

Reordering	Linear
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Resolution - Common

FOV Read	240 mm
FOV Phase	93.8 %
Slice Thickness	0.63 mm
Base Resolution	384
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	7.3
Reference Scans	GRE/Separate
Acceleration Factor PE	1
Reference Lines PE	32
Acceleration Factor 3D	1
Reordering Shift 3D	0
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	B1 Filter
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A12.2 F22.8 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	256
Phase Oversampling	0 %
Slice Oversampling	12.5 %
FOV Read	240 mm
FOV Phase	93.8 %
Slice Thickness	0.63 mm
TR	6000.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A12.2 F22.8 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 A12.2 F22.8
R	0.0 mm
A	12.2 mm
F	22.8 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Tim Planning Suite**

Set-n-Go Protocol	Off
Table Position	23 mm
Table Position	F

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	L >> R
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	Patient-specific
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 A12.2 F22.8 mm
Orientation	Sagittal
Rotation	0.00 deg
A >> P	225 mm
F >> H	240 mm
R >> L	162 mm
Reset	Off

System - pTx

B1 Shim	Patient-specific
Excitation	Non-sel.
pTx Pulse	1

System - Tx/Rx

Frequency 1H	297.208891 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	6000.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	Non-sel. IR
TI 1	875 ms
TI 2	2700 ms
Dark Blood	Off
FOV Read	240 mm
FOV Phase	93.8 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - MapIt**

MapIt	T1 Map
Flip Angle 1	4 deg
Flip Angle 2	5 deg
Measurements	1
Contrasts	1
TE	3.02 ms
TR	6000.0 ms
Save Original Images	On

Sequence - Part 1

Sequence Name	WIP_tfl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	240 Hz/Px
Echo Spacing	7.12 ms
Asymmetric Echo	Off
Turbo Factor	205

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Special

Interpolation Mode	Post Volume
DL Recon	DL-Finetuned_CS-7
Run in Background	Off
Data Fidelity	3.0

Sequence - Assistant

SAR Assistant	Off
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\\USER\Renzo\PTx_protocols\copied\PTX_SAG_DIR_FLAIR_SPC_Q9WS104-0.78mm

TA: 7:06 min Coil Selection: Auto Voxel Size: 0.8×0.8×0.8 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	1st Segment
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	L0.0 A9.1 F24.2 mm
Orientation	S > T-0.3 > C-0.1
Phase Encoding Dir.	A >> P
Slices per Slab	208
Phase Oversampling	0 %
Slice Oversampling	7.7 %
FOV Read	225 mm
FOV Phase	100.0 %
Slice Thickness	0.78 mm
TR	8000.0 ms
TE	302.00 ms
Averages	1.0
Concatenations	1
AutoAlign	Head > Brain

Contrast - Common

TR	8000.0 ms
TE	302.00 ms
Magn. Preparation	Non-sel. DIR
TI 1	3120 ms
TI 2	450 ms
Flip Angle Mode	T2 Var
Fat-Water Contrast	Fat Saturation
Fat Saturation	Strong
Dark Blood	Off
Blood Suppression	Off
Excitation offset frequency	0 Hz
Wrap-up Magn.	None
Refoc. offset frequency	0 Hz
Reconstruction	Magnitude

Contrast - Dynamic

goKART	Off
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Contrast - Dynamic

Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Contrast - ASL

MTC	Off
MTC frequency sign	Positive

Resolution - Common

FOV Read	225 mm
FOV Phase	100.0 %
Slice Thickness	0.78 mm
Base Resolution	288
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Total Factor	6
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	32
Acceleration Factor 3D	2
Reference Lines 3D	32
Reordering Shift 3D	1
Deep Resolve	Boost
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Elliptical Scanning	On

Resolution - Filter

Raw Filter	On
Elliptical Filter	Off
Distortion Correction	3D
Normalize	B1 Filter
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 A9.1 F24.2 mm
Orientation	S > T-0.3 > C-0.1
Phase Encoding Dir.	A >> P
Slices per Slab	208
Phase Oversampling	0 %
Slice Oversampling	7.7 %
FOV Read	225 mm
FOV Phase	100.0 %
Slice Thickness	0.78 mm

Geometry - Common

TR	8000.0 ms
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A9.1 F24.2 mm
Orientation	S > T-0.3 > C-0.1
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 A15.7 F20.3
R	0.0 mm
A	15.7 mm
F	20.3 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	L >> R
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Always
Assume Silicone	Off
Adj Freq. protocol related	On

System - Adjust Volume

Position	L0.0 A9.1 F24.2 mm
Orientation	S > T-0.3 > C-0.1
Rotation	10.43 deg
A >> P	225 mm
F >> H	225 mm
R >> L	163 mm

System - Adjust Volume

Reset	Off
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System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.
pTx Pulse	1

System - Tx/Rx

Frequency 1H	297.208891 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000
Gain	High

Physio - Signal

1st Signal/Mode	None
Trigger Delay	0 ms
TR	8000.0 ms
Rd crusher scale	893
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fat Saturation
Magn. Preparation	Non-sel. DIR
TI 1	3120 ms
TI 2	450 ms
Dark Blood	Off
FOV Read	225 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Sequence - Part 1

Sequence Name	wip-spc-dir
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Low SAR
Refoc. duration	1000
Gradient Mode	Normal
Flow Compensation	None
Reordering	Linear
Bandwidth	755 Hz/Px
Echo Spacing	4.14 ms
Concomitant field corr. R	Product
Turbo Factor	180
Echo Train Duration	741 ms
Rd crusher scale	893

Sequence - Part 2

Introduction	On
Rel. crusher mom. P	0 %
Rel crusher mom. S	0 %

Sequence - Special

IR refoc pulse	IR12800H180.IR180_HS3
IR RF duration	12800 us
IR HS mu	5.00
IR HS BWTP	10.99
IR HS B1 peak	22.5 μ T
Excit. flip angle	90 deg
Blood Suppression	Off
Trajectory Optimization	8

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

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TA: 7:06 min Coil Selection: Auto Voxel Size: 0.8x0.8x0.8 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	1st Segment
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	L0.0 P3.9 F29.8 mm
Orientation	S > T-0.3 > C-0.1
Phase Encoding Dir.	A >> P
Slices per Slab	208
Phase Oversampling	0 %
Slice Oversampling	7.7 %
FOV Read	225 mm
FOV Phase	100.0 %
Slice Thickness	0.78 mm
TR	8000.0 ms
TE	302.00 ms
Averages	1.0
Concatenations	1
AutoAlign	Head > Brain

Contrast - Common

TR	8000.0 ms
TE	302.00 ms
Magn. Preparation	Non-sel. T2 prep. IR
T1	2250 ms
T2 Prep. Duration	125 ms
Flip Angle Mode	T2 Var
Fat-Water Contrast	Fat Saturation
Fat Saturation	Strong
Dark Blood	Off
Blood Suppression	Off
Excitation offset frequency	0 Hz
Wrap-up Magn.	None
Refoc. offset frequency	0 Hz
Reconstruction	Magnitude

Contrast - Dynamic

goKART	Off
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Contrast - Dynamic

Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Contrast - ASL

MTC	Off
MTC frequency sign	Positive

Resolution - Common

FOV Read	225 mm
FOV Phase	100.0 %
Slice Thickness	0.78 mm
Base Resolution	288
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Total Factor	6
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	32
Acceleration Factor 3D	2
Reference Lines 3D	32
Reordering Shift 3D	1
Deep Resolve	Boost
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Elliptical Scanning	On

Resolution - Filter

Raw Filter	On
Elliptical Filter	Off
Distortion Correction	3D
Normalize	B1 Filter
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 P3.9 F29.8 mm
Orientation	S > T-0.3 > C-0.1
Phase Encoding Dir.	A >> P
Slices per Slab	208
Phase Oversampling	0 %
Slice Oversampling	7.7 %
FOV Read	225 mm
FOV Phase	100.0 %
Slice Thickness	0.78 mm

Geometry - Common

TR	8000.0 ms
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 P3.9 F29.8 mm
Orientation	S > T-0.3 > C-0.1
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 A2.0 F23.4
R	0.0 mm
A	2.0 mm
F	23.4 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	L >> R
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Always
Assume Silicone	Off
Adj Freq. protocol related	On

System - Adjust Volume

Position	L0.0 P3.9 F29.8 mm
Orientation	S > T-0.3 > C-0.1
Rotation	10.43 deg
A >> P	225 mm
F >> H	225 mm
R >> L	163 mm

System - Adjust Volume

Reset	Off
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System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.
pTx Pulse	1

System - Tx/Rx

Frequency 1H	297.208891 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000
Gain	High

Physio - Signal

1st Signal/Mode	None
Trigger Delay	0 ms
TR	8000.0 ms
Rd crusher scale	893
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fat Saturation
Magn. Preparation	Non-sel. T2 prep. IR
TI	2250 ms
T2 Prep. Duration	125 ms
Dark Blood	Off
FOV Read	225 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Sequence - Part 1

Sequence Name	wip-spc-t2p+ir
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Low SAR
Refoc. duration	1000
Gradient Mode	Normal
Flow Compensation	None
Reordering	Linear
Bandwidth	755 Hz/Px
Echo Spacing	4.14 ms
Concomitant field corr. R	Product
Turbo Factor	180
Echo Train Duration	741 ms
Rd crusher scale	893

Sequence - Part 2

Introduction	On
Rel. crusher mom. P	0 %
Rel crusher mom. S	0 %

Sequence - Special

IR refoc pulse	IR12800H180.IR180_HS3
IR RF duration	12800 us
IR HS mu	5.00
IR HS BWTP	10.99
IR HS B1 peak	22.5 μ T
T2-Prep. excit/tip-up pulse	RECT
T2-Prep. refoc pulse	BIREF-1
T2-Prep. # refoc. RF	4
T2-Prep. excit. duration	800 us
T2-Prep. refoc. duration	11520 us
T2-Prep. spoil mom. 10^3 x	-71.20 usmT/m
T2-Prep. B1 scale	1.00
T2-Prep. crusher mom. rel.	0.00
T2-Prep. Refoc shift	0.00 %
Excit. flip angle	90 deg
Blood Suppression	Off
Trajectory Optimization	8

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

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TA: 7:08 min Coil Selection: Auto Voxel Size: 0.3x0.3x0.6 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	1st Segment
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	R0.1 A8.0 F2.7 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	288
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	230 mm
FOV Phase	100.0 %
Slice Thickness	0.60 mm
TR	2650.0 ms
TE	141.00 ms
Averages	1.0
Concatenations	1
AutoAlign	Head > Brain

Contrast - Common

TR	2650.0 ms
TE	141.00 ms
Magn. Preparation	None
Flip Angle Mode	T2 Var
Fat-Water Contrast	Standard
Dark Blood	Off
Blood Suppression	Off
Excitation offset frequency	0 Hz
Wrap-up Magn.	None
Refoc. offset frequency	0 Hz
Reconstruction	Magnitude

Contrast - Dynamic

goKART	Off
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Contrast - ASL

MTC	Off
MTC frequency sign	Positive

Resolution - Common

FOV Read	230 mm
FOV Phase	100.0 %
Slice Thickness	0.60 mm
Base Resolution	384
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	On

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Total Factor	6
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	32
Acceleration Factor 3D	2
Reference Lines 3D	32
Reordering Shift 3D	1
Deep Resolve	Boost
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Elliptical Scanning	On

Resolution - Filter

Raw Filter	On
Elliptical Filter	Off
Distortion Correction	3D
Normalize	B1 Filter
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	R0.1 A8.0 F2.7 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	288
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	230 mm
FOV Phase	100.0 %
Slice Thickness	0.60 mm
TR	2650.0 ms
Concatenations	1

Geometry - AutoAlign

Slab Group	1
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Geometry - AutoAlign

Position	R0.1 A8.0 F2.7 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	R0.1 A8.0 F2.7
R	0.1 mm
A	8.0 mm
F	2.7 mm
Initial Orientation	Sagittal
Initial Rotation	-0.76 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	3 mm
Table Position	F

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	L >> R
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj Freq. protocol related	Off

System - Adjust Volume

Position	R0.1 A8.0 F2.7 mm
Orientation	Sagittal
Rotation	-0.76 deg
A >> P	230 mm
F >> H	230 mm
R >> L	173 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - pTx

pTx Pulse	1
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System - Tx/Rx

Frequency 1H	297.208891 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000
Gain	High

Physio - Signal

1st Signal/Mode	None
Trigger Delay	0 ms
TR	2650.0 ms
Rd crusher scale	893
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	230 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Sequence - Part 1**

Sequence Name	wip-spc
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Low SAR
Refoc. duration	1000
Gradient Mode	Normal

Sequence - Part 1

Flow Compensation	None
Reordering	Linear
Bandwidth	521 Hz/Px
Echo Spacing	4.54 ms
Concomitant field corr. R	Product
Turbo Factor	96
Echo Train Duration	431 ms
Rd crusher scale	893

Sequence - Part 2

Introduction	On
Rel. crusher mom. P	0 %
Rel crusher mom. S	0 %

Sequence - Special

Excit. flip angle	90 deg
Blood Suppression	Off
Trajectory Optimization	8

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

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TA: 6:03 min Coil Selection: Auto Voxel Size: 0.6×0.6×0.6 mm³ Acc:: 6.7 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	On
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	1st Segment
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A6.3 F22.3 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	256
Phase Oversampling	0 %
Slice Oversampling	12.5 %
FOV Read	240 mm
FOV Phase	93.8 %
Slice Thickness	0.63 mm
TR	6000.0 ms
TE	3.02 ms
Averages	1
Concatenations	1
AutoAlign	Head > Brain

Contrast - Common

TR	6000.0 ms
TE	3.02 ms
Magn. Preparation	Non-sel. IR
TI 1	875 ms
TI 2	2700 ms
Flip Angle 1	4 deg
Flip Angle 2	5 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Contrast - Dynamic

Reordering	Linear
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Resolution - Common

FOV Read	240 mm
FOV Phase	93.8 %
Slice Thickness	0.63 mm
Base Resolution	384
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	6.7
Reference Scans	GRE/Separate
Reference Lines PE	32
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	B1 Filter
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A6.3 F22.3 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	256
Phase Oversampling	0 %
Slice Oversampling	12.5 %
FOV Read	240 mm
FOV Phase	93.8 %
Slice Thickness	0.63 mm
TR	6000.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A6.3 F22.3 mm
Orientation	Sagittal

Geometry - AutoAlign

Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 A6.3 F22.3
R	0.0 mm
A	6.3 mm
F	22.3 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Tim Planning Suite**

Set-n-Go Protocol	Off
Table Position	22 mm
Table Position	F

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	L >> R
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 A6.3 F22.3 mm
Orientation	Sagittal
Rotation	0.00 deg
A >> P	225 mm
F >> H	240 mm
R >> L	162 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.
pTx Pulse	1

System - Tx/Rx

Frequency 1H	297.208891 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off

System - Tx/Rx

Image Scaling	1.000
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Physio - Signal

1st Signal/Mode	None
TR	6000.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	Non-sel. IR
TI 1	875 ms
TI 2	2700 ms
Dark Blood	Off
FOV Read	240 mm
FOV Phase	93.8 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - MapIt**

MapIt	T1 Map
Flip Angle 1	4 deg
Flip Angle 2	5 deg
Measurements	1
Contrasts	1
TE	3.02 ms
TR	6000.0 ms
Save Original Images	On

Sequence - Part 1

Sequence Name	tfl
Dimension	3D
Excitation	Non-sel.

Sequence - Part 1

RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	240 Hz/Px
Echo Spacing	7.12 ms
Asymmetric Echo	Off
Turbo Factor	200

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Assistant

SAR Assistant	Off
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\\USER\Renzo\PTx_protocols\copied\COR T2 TSE HIPPO DR +PSS

TA: 7:59 min Coil Selection: Auto Voxel Size: 0.2x0.2x1.2 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	On
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	2nd Segment
Inline Movie	Off

Resolution - Common

FOV Read	150 mm
FOV Phase	100.0 %
Slice Thickness	1.2 mm
Base Resolution	448
Phase Resolution	100 %
Interpolation	On

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	TSE/Separate
Acceleration Factor PE	2
Reference Lines PE	32
Deep Resolve	On
Phase Partial Fourier	Off

Routine

Slice Group	1
Slices	60
Distance Factor	20 %
Position	L1.1 A10.3 H9.3 mm
Orientation	C > T-5.1 > S0.7
Phase Encoding Dir.	R >> L
Phase Oversampling	24 %
FOV Read	150 mm
FOV Phase	100.0 %
Slice Thickness	1.2 mm
TR	8180.0 ms
TE	31.00 ms
Averages	2
Concatenations	1
AutoAlign	Head > Temporal Lobe

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	B1 Filter
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	60
Distance Factor	20 %
Position	L1.1 A10.3 H9.3 mm
Orientation	C > T-5.1 > S0.7
Phase Encoding Dir.	R >> L
Phase Oversampling	24 %
FOV Read	150 mm
FOV Phase	100.0 %
Slice Thickness	1.2 mm
TR	8180.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Contrast - Common

TR	8180.0 ms
TE	31.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	110 deg
Fat-Water Contrast	Fat Saturation
Fat Saturation	Weak
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Geometry - AutoAlign

Slice Group	1
Position	L1.1 A10.3 H9.3 mm
Orientation	C > T-5.1 > S0.7
Phase Encoding Dir.	R >> L
AutoAlign	Head > Temporal Lobe
Initial Position	L1.1 A10.3 H9.3
L	1.1 mm
A	10.3 mm
H	9.3 mm
Initial Orientation	C > T
C > T	-5.10

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Geometry - AutoAlign

> S	0.70
Initial Rotation	-1.21 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	9 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	L >> R
Coronal	P >> A
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	Patient-specific
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L1.1 A10.3 H9.3 mm
Orientation	C > T-5.1 > S0.7
Rotation	-1.21 deg
R >> L	150 mm
F >> H	150 mm
A >> P	87 mm
Reset	Off

System - pTx

B1 Shim	Patient-specific
LR Balancing	Off

System - Tx/Rx

Frequency 1H	297.208891 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
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Physio - Signal

TR	8180.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fat Saturation
Magn. Preparation	None
Dark Blood	Off
FOV Read	150 mm
FOV Phase	100.0 %
Phase Resolution	100 %
Motion Correction	None

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Motion Correction	None
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Sequence - Part 1**

Sequence Name	tse_rr
Dimension	2D
RF Pulse Type	Low SAR
Gradient Mode	Fast*
Flow Compensation	Read
Bandwidth	272 Hz/Px
Echo Spacing	10.4 ms
Free Echo Spacing	Off
Define	Turbo Factor
Turbo Factor	10
Echo Trains per Slice	28

Sequence - Part 2

Introduction	Off
Incr. Gradient Spoiling	Off
Phase Correction	Off
Compensate T2 Decay	Off

Sequence - Part 2

Fast Mode	Off
WARP	Off
Acoustic noise reduction	Off
Reduce Motion Sens.	On
Motion Correction	None

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	60 s